

Algebra 1
Unit 2
Lesson 1 Homework

Name Answer Key
Date _____
Period _____

1. The product of powers law of exponents states to keep the then _____ the powers.

☐ add
☐ subtract
☐ multiply

☐ base
☐ coefficient and
☐ exponent

2. Jeffrey simplified the algebraic expression and made an error. Find and fix the mistake.

$$\begin{aligned} &(-2x^2y^3)^4(3.5x^0y) \\ &\quad (-2x^8y^{12})(3.5y) \\ &(-2 \cdot 3.5)(x^8)(y^{12} \cdot y) \\ &\quad -7.5x^8y^{13} \end{aligned}$$

$$\begin{aligned} &(-2x^2y^3)^4(3.5x^0y) \\ &\quad (16x^8y^{12})(3.5y) \\ &\quad 56x^8y^{13} \end{aligned}$$

3. Write a note to Jeffrey explaining the error he made in the previous question.

Apply the Power of a Product Law of Exponents to the numerical coefficients.

Determine an equivalent algebraic monomial expression for each expression using the laws of exponents. Show your work.

4. $(-4a^2b)(5ab^3)$

$$-20a^3b^4$$

5. $(\frac{-1}{2}x^3y^0)(6x^2y^3)(-3y)$

$$9x^5y^4$$

$$6. \quad (5.6kj^8)(-3.25k^9)(\frac{1}{4}k^7j^{12})$$

$$-\frac{91}{20}k^{17}j^{20} \quad OR \quad -4.55k^{17}j^{20}$$

$$7. \quad (-5mn^3p^5)^2$$

$$25m^2n^6p^{10}$$

$$8. \quad (4.1c^{-4}d)(-2.2c^{10}d^0)(-c^{-2}d^3)$$

$$9.02c^4d^4$$

$$9. \quad (\frac{5}{8}r^{-2})^0(-7.6r^4s^5)^{-2}$$

$$\frac{25}{1444r^8s^{10}} \quad OR \quad \frac{1}{57.76r^8s^{10}}$$

$$10. \quad (-2y^3z)^5(6y^{10}z^8)^2$$

$$-1152y^{35}z^{21}$$

Algebra 1
Unit 2
Lesson 2 Homework

Name Answer Key
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1. The **quotient of powers law** of exponents states keep the then _____ the powers.

add

subtract

multiply

base

coefficient

exponent

and

2. The **power of a quotient law** of exponents states the exponential expression $\left(\frac{a}{b}\right)^m$ is equal to

☐ a^{b-m} .

☒ $\frac{a^m}{b^m}$.

☐ $a^m - b^m$.

Find the quotient of each expression by using the laws of exponents to generate an equivalent expression.

3. $\frac{-4x^4y^2}{10xy^7}$

$\frac{-2x^3}{5y^5}$

4. $\frac{3a^{10}b^6}{6b^2c}$

$\frac{a^{10}b^4}{2c}$

5. $\left(\frac{-2r^3s}{3t^4}\right)^6$

$\frac{64r^{18}s^6}{729t^{24}}$

6. $\frac{-k^7j^3}{7k^9j^3}$

$-\frac{1}{7k^2}$

$$7. \quad \frac{46w^3z^{10}}{33x^2y^{11}}$$

$$\frac{46w^3z^{10}}{33x^2y^{11}}$$

$$8. \quad \frac{-144c^9d^2e^0}{98c^4d^{11}e}$$

$$\frac{-72c^5}{49d^9e}$$

$$9. \quad \left(\frac{z^0}{-3x^3}\right)^4$$

$$\frac{1}{81x^{12}}$$

$$10. \quad \frac{5s^{15}t^6}{-10s^{12}t}$$

$$-\frac{s^3t^5}{2}$$

Algebra 1
Unit 2
Lesson 3 Homework

Name Answer Key
Date _____
Period _____

1. Tamer was simplifying the expression and made an error. His steps are shown. Find and fix the mistake.

$$\left(\frac{-2xy \cdot 3x^2}{5x^4y}\right)^2$$

Step 1: $\left(\frac{-6x^3y}{5x^4y}\right)^2$

Step 2: $\frac{-6x^6y^2}{5x^8y^2}$

Step 3: $\frac{-6}{5x^2}$

Step 1: $\left(\frac{-6x^3y}{5x^4y}\right)^2$

Step 2: $\frac{36x^6y^2}{25x^8y^2}$

Step 3: $\frac{36}{25x^2}$

2. Write a note to Tamer explaining his error.

Apply Power of a Product Law of Exponents also to numerical coefficients.

For questions 3-5, find an equivalent expression using the laws of exponents.

3. $\frac{-6a^7b^9}{(-2ab^3)^5}$

$$\frac{-6a^7b^9}{-32a^5b^{15}} = \frac{3a^2}{16b^6}$$

4. $\frac{(9x^8y) \cdot (-2x^2y^5)}{3x^5y^{10}}$

$$\frac{-18x^{10}y^6}{3x^5y^{10}} = \frac{-6x^5}{y^4}$$

$$5. \quad \left(\frac{-3mn^3}{4m^4 \cdot 2m^2n^5}\right)^3$$

$$\frac{-27m^3n^9}{512m^{18}n^{15}} = \frac{-27}{512m^{15}n^6}$$

$$6. \quad \text{Prove } 2^{3x+5} = 32(8)^x \text{ by using the laws of exponents.}$$

$$\begin{aligned} 2^{3x+5} &= 32(8)^x \\ (2^{3x})(2^5) &= 32(8)^x \\ 32(2^3)^x &= 32(8)^x \\ 32(8)^x &= 32(8)^x \end{aligned}$$

$$7. \quad \text{Prove } 27^{6a-1} = \frac{(3^{6a})^3}{9 \cdot 3} \text{ by using the laws of exponents.}$$

$$\begin{aligned} 27^{6a-1} &= \frac{(3^{6a})^3}{9 \cdot 3} \\ (27^{6a})(27^{-1}) &= \frac{(3^{6a})^3}{9 \cdot 3} \\ [(3^3)^{6a}](27^{-1}) &= \frac{(3^{6a})^3}{9 \cdot 3} \\ \frac{3^{18a}}{27} &= \frac{(3^{6a})^3}{9 \cdot 3} \\ \frac{(3^{6a})^3}{9 \cdot 3} &= \frac{(3^{6a})^3}{9 \cdot 3} \end{aligned}$$

$$8. \quad \text{Write an equivalent expression for } (-5.2)^{2x^3} \cdot (-5.2)^{x^2} \cdot (-5.2)^{-5}.$$

$$(-5.2)^{2x^3+x^2-5}$$

$$9. \quad \text{Use the law of exponents to write an equivalent expression for } 4^{3-2c}.$$

$$(4^3)(4^{-2c}) = \frac{4^3}{4^{2c}}$$

$$10. \quad \text{Use the law of exponents to write an equivalent expression for } 3.5^{3x+7}.$$

$$(3.5^{3x})(3.5^7) = (3.5^x)^3(3.5^7) = (3.5^3)^x(3.5^7)$$

Algebra 1
Unit 2
Lesson 4 Homework

Name Answer Key
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1. True or False? The polynomial $-7xy^4 + 4x^5y^2 - 8.6x^3y^3$ is written in standard form.

False

2. Write the polynomial $-7xy^4 + 4x^5y^2 - 8.6x^3y^3$ in standard form.

$$4x^5y^2 - 8.6x^3y^3 - 7xy^4$$

For questions 3-4, rewrite each polynomial in standard form, then identify the features of the polynomial.

3. $2.5y^3 + 7y - 8\frac{7}{8}y^5$

Standard Form: $-8\frac{7}{8}y^5 + 2.5y^3 + 7y$

Number of Terms: 3

Degree of the polynomial: 5

Leading Term: $-8\frac{7}{8}y^5$

Leading Coefficient: $-8\frac{7}{8}$

4. $-\frac{3}{4}a^4b^2 - 16ab^3 + ab^6$

Standard Form: $ab^6 - \frac{3}{4}a^4b^2 - 16ab^3$

Number of Terms: 3

Degree of the polynomial: 7

Leading Term: ab^6

Leading Coefficient: 1

5. Create a polynomial that has 4 terms, a degree of 6, a leading coefficient of -1 , and a constant term of -5.5 .

$$-x^6 + 2x^5 - 3x^2 - 5.5 \quad \text{Answers will vary.}$$

6. Samantha found the product of $-2g^3(\frac{1}{4}g^8h^3 + 5g^2h^6 - g^9)$ to be $-\frac{1}{2}g^{24}h^3 - 10g^6h^6 - 2g^{27}$. Do you agree or disagree with her? Explain.

Disagree. She did not apply the law of exponents correctly.

7. Find the product of $-2g^3(\frac{1}{4}g^8h^3 + 5g^2h^6 - g^9)$ in standard form.

$$-\frac{1}{2}g^{11}h^3 - 10g^5h^6 + 2g^{12}$$

$$-\frac{1}{2}g^{11}h^3 + 2g^{12} - 10g^5h^6$$

Find the product of each by distributing the monomial to the polynomial. Express the product in standard form.

8. $-5x^2(-3xy + 2x^7 - 7x^3)$

$$15x^3y - 10x^9 + 35x^5$$

$$-10x^9 + 35x^5 + 15x^3y$$

9. $(-\frac{5}{8}c^4d + 10c^5d^3 - cd^2)(\frac{3}{5}c^2)$

$$-\frac{3}{8}c^6d + 6c^7d^3 - \frac{3}{5}c^3d^2$$

$$6c^7d^3 - \frac{3}{8}c^6d - \frac{3}{5}c^3d^2$$

10. $-4.2a^3(2b^5 - 1.5a^2b - 9)$

$$-8.4a^3b^5 + 6.3a^5b + 37.8a^3$$

Algebra 1
Unit 2
Lesson 5 Homework

Name Answer Key
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1. Are the terms $-4xy^2$ and $\frac{1}{2}x^2y$ like terms? Why or why not? Explain.

No, exponents of each variable are different.

2. Find the sum of $(6 - t - t^2)$ and $(10t + t^2)$.

$$(6 - t - t^2) + (10t + t^2) = 9t + 6$$

3. Write the sum from question 2 in standard form.

$$9t + 6$$

4. Find the difference of $(5.5n^2 + 8 - 3n)$ and $(-3n^2 - 5)$.

$$(5.5n^2 + 8 - 3n) - (-3n^2 - 5)$$

$$8.5n^2 + 13 - 3n$$

5. Write the difference from question 4 in standard form.

$$8.5n^2 - 3n + 13$$

6. What polynomial must be added to $-5c^2 - 3 + 4c$ in order to get a sum of $-2c^2 - c + 1$?

$$(-2c^2 - c + 1) - (-5c^2 - 3 + 4c)$$

$$3c^2 - 5c + 4$$

For questions 7-10, find each sum or difference. Write the result in standard form.

7. $(-3.4g^2h - 5gh^2 + 7) + (-1.75gh^2 - 12)$

$$-3.4g^2h - 6.75gh^2 - 5$$

8. $2\frac{1}{5}r^5t^2 - (3r^4t^3 + r^5t^2 - 6rt^4)$

$$\frac{11}{5}r^5t^2 - 3r^4t^3 - r^5t^2 + 6rt^4$$

$$\frac{6}{5}r^5t^2 - 3r^4t^3 + 6rt^4$$

9. $(8a^2 - 3a + 11b^2) + (2a - 5b^2 - 2a^2)$

$$6a^2 + 6b^2 - a$$

10. $(9.3m^2 - 5.1n) - (2n^2 + 6.75m)$

$$9.3m^2 - 2n^2 - 6.75m - 5.1n$$

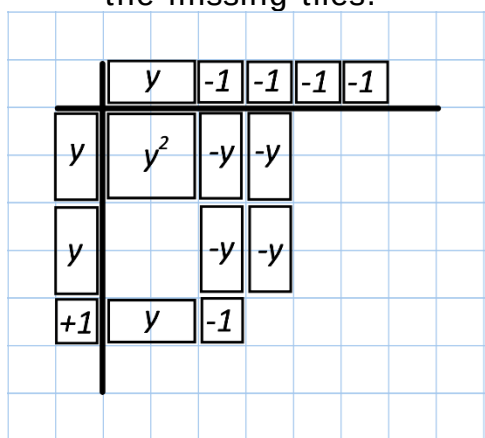
1. Model the following expression with algebra tiles.

$$5x - 4$$

x	x	x	x	x	-1	-1	-1	-1
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Use the expression $(2y + 1)(y - 4)$ for questions 2 – 4.

2. Complete the model to find the product of $(2y + 1)(y - 4)$ by drawing and labeling the missing tiles.



	y	-1	-1	-1	-1
y	y^2	$-y$	$-y$	$-y$	$-y$
y	y^2	$-y$	$-y$	$-y$	$-y$
+1	y	-1	-1	-1	-1

3. Determine how many of each tile are represented in the product and write the numbers in the table.

y^2 tiles	$+y$ tiles	$-y$ tiles	$+1$ tiles	-1 tiles
2	1	8	0	4

4. Complete the statement.

The product of $(2y + 1)(y - 4)$ is $2y^2 - 7y - 4$

For questions 5-9, use algebra tiles to find each product.

5. $(4 - 3x)(x - 1) = -3x^2 + 7x - 4$

	-x	-x	-x	1	1	1	1
x	$-x^2$	$-x^2$	$-x^2$	x	x	x	x
-1	x	x	x	-1	-1	-1	-1

6. $(2x - 1)^2 = 4x^2 - 4x + 1$

	x	x	-1
x	x^2	x^2	$-x$
x	x^2	x^2	$-x$
-1	$-x$	$-x$	1

7. $(x - 3)(x + 3) = x^2 - 9$

	x	-1	-1	-1
x	x^2	$-x$	$-x$	$-x$
+1	x	-1	-1	-1
+1	x	-1	-1	-1
+1	x	-1	-1	-1

8. $(x - 5)(x + 1) = x^2 - 4x - 5$

	x	-1	-1	-1	-1	-1
x	x^2	$-x$	$-x$	$-x$	$-x$	$-x$
1	x	-1	-1	-1	-1	-1

9. $(1 - x)(2x + 2) = -2x^2 + 2$

	-x	+1
x	$-x^2$	x
x	$-x^2$	x
+1	$-x$	+1
+1	$-x$	+1

10. One of your classmates was absent when you learned how to multiply polynomials using algebra tiles. Write them a note explaining the process.

Multiplying polynomials using algebra tiles is just like finding the area of a rectangle. Line-up the factors to represent length and width. Fill in the area with blocks that are the same length and width. The blocks in the area space represent the product of the polynomials.

Algebra 1
Unit 2
Lesson 7 Homework

Name Answer Key
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1. Find the product of $(x^2 - 3)(4x + 2)$ using the distributive property.

$$4x^3 + 2x^2 - 12x - 6$$

2. Find the product of $(x^2 - 3)(4x + 2)$ using an area model.

	x^2	-3
$4x$	$4x^3$	$-12x$
$+2$	$2x^2$	-6

$$4x^3 + 2x^2 - 12x - 6$$

3. Find the product of $(-\frac{1}{4}y + 3y^2 - 7)(2y - 5)$ using the distributive property.

$$-\frac{1}{2}y^2 + \frac{5}{4}y + 6y^3 - 15y^2 - 14y + 35$$

$$6y^3 - 15\frac{1}{2}y^2 - 12\frac{3}{4}y + 35$$

4. Find the product of $(-\frac{1}{4}y + 3y^2 - 7)(2y - 5)$ using an area model.

	$-\frac{1}{4}y$	$3y^2$	-7
$2y$	$-\frac{1}{2}y^2$	$6y^2$	$-14y$
-5	$\frac{5}{4}y$	$-15y^2$	35

$$6y^3 - 15\frac{1}{2}y^2 - 12\frac{3}{4}y + 35$$

Find each product.

5. $(-6a + \frac{2}{5}b)^2$

$$36a^2 - \frac{24}{5}ab + \frac{4}{25}b^2$$

6. $(2.4 - 3c)(-0.3 + 5c^3 - c^2)$

$$-15c^4 + 15c^3 - 2.4c^2 + 0.9c - 0.72$$

7. $(\frac{3}{4}x - 7)(\frac{-2}{5} + 8x)$

$$6x^2 - 56.3x + 2.8$$

8. $(-2\frac{1}{2} + 5d^2)(6d - 6.1c - 3d^2)$

$$-15d^4 + 30d^3 + 7.5d^2 - 30cd^2 + 15.25c - 15d$$

9. $(12 + 6m)(-8n - \frac{18}{5})$

$$-48mn - 21.6m - 96n - 43.2$$

10. $(4r^2 - \frac{7}{4} + 19r)(-3r + 2r^2 + 15)$

$$8r^4 + 26r^3 - 0.5r^2 + 290.25r - 26.25$$

Algebra 1
Unit 2
Lesson 8 Homework

Name Answer Key
Date
Period

1. Rewrite the expression in each step using the description: $-2(x - 2) + 3x$

$-2x + 4$	Distribute the -2 to $(x - 2)$.
$-2x + 4 + 3x$ $x + 4$	Combine like terms.

2. Rewrite the expression in each step using the description: $5x + 6(2x + 7)$

$12x + 42$	Distribute the 6 to $(2x + 7)$.
$5x + 12x + 42$ $17x + 42$	Combine like terms.

Use the order of operations to simplify each polynomial.

3. $4(x - 3) + 6(x - 2)$

$$4x - 12 + 6x - 12$$

$$10x - 24$$

4. $\frac{2}{3}(x - 3) + 6x$

$$\frac{2}{3}x - 2 + 6x$$

$$6\frac{2}{3}x - 2$$

5. $2(x^2 - 3x) + 4(x + 5)$

$$2x^2 - 6x + 4x + 20$$

$$2x^2 - 2x + 20$$

6. $4x^2 - 4(x - 8)$

$$4x^2 - 4x + 32$$

7. $(x + 2)(x - 3) - 6(x^2 + x)$

$$x^2 - 3x + 2x - 6 - 6x^2 - 6x$$

$$-5x^2 - 7x - 6$$

8. $-\frac{2}{5}(5x - 15) + 5x$

$$-2x + 6 + 5x$$

$$3x + 6$$

9. $x^2 + (x - 2)(x - 2)$

$$x^2 + x^2 - 2x - 2x + 4$$

$$2x^2 - 4x + 4$$

10. $4x(x + 1) + 2(x + 5)$

$$4x^2 + 4x + 2x + 10$$

$$4x^2 + 6x + 10$$

Algebra 1
Unit 2
Lesson 9 Homework

Name Answer Key
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1. Use the area model to find $(-4.5x^4y^3 + 2.4x^3y^2 - 3.6x^2y) \div (3xy)$ where $x \neq 0$ and $y \neq 0$.

$$-1.5x^3y^2 + 0.8x^2y - 1.2x$$

	$-1.5x^3y^2$	$0.8x^2y$	$-1.2x$
$3xy$	$-4.5x^4y^3$	$2.4x^3y^2$	$-3.6x^2y$

Find each quotient.

2. $(\frac{3}{4}x^3 + \frac{1}{2}x^2) \div (-\frac{1}{2}x^2)$ where $x \neq 0$.

$$-1.5x - 1$$

3. $(4a^3 + 8a^2 - 20a) \div (4a)$ where $a \neq 0$

$$a^2 + 2a - 5$$

4. $(-10k^6r^4 + 6k^2r^4 - 42k^3r^3) \div (2k^2r)$ where $k \neq 0$ and $r \neq 0$

$$-5k^4r^3 + 3r^3 - 21kr^2$$

5. $(-17w^6 + 13w^3 - 18w^2) \div (2w)$ where $w \neq 0$

$$-8.5w^5 + 6.5w^2 - 9w$$

6. $\frac{12x^4 - 30x^3 + 6x^2 - 18}{6x}$ where $x \neq 0$

$$2x^3 - 5x^2 + x - \frac{3}{x}$$

7. $\frac{14p^4+42p^3+21p^2-7p}{7p}$ where $p \neq 0$

$$2p^3 + 6p^2 + 3p - 1$$

8. $\frac{2t^4-2t^3+12t^2-7t}{2t}$ where $t \neq 0$

$$t^3 - t^2 + 6t - \frac{7}{2}$$

9. $\frac{2.8x^3+4.8+3.2x^2-0.8}{0.4x}$ where $x \neq 0$

$$7x^2 + \frac{10}{x} + 8x$$

10. $\frac{15y^3+45y^2+35y^2}{-5y^2}$ where $y \neq 0$

$$-3y - 16$$